EMCH 792: Optimal State Estimation
Homework 3

Written Portion Solutions

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Due: February 26, 2026

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Exercise 2.5 (Page 75): Exponentially Distributed Random Variable

Given

$$f_X(x) = \begin{cases} \alpha e^{-\alpha x}, & x > 0, \\ 0, & x \le 0, \end{cases} \qquad \alpha > 0.$$

Derivation

(a) Find the CDF $F_X(x)$

By definition,

$$F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(z)\, dz.$$

Because the density is zero for $z \le 0$, evaluate in two regions:

$$F_X(x) = 0, \qquad x \le 0.$$

For $x > 0$,

$$F_X(x) = \int_0^x \alpha e^{-\alpha z}\, dz = \left[-e^{-\alpha z}\right]_0^x = 1 - e^{-\alpha x}.$$

Therefore,

$$F_X(x) = \begin{cases} 0, & x \le 0, \\ 1 - e^{-\alpha x}, & x > 0. \end{cases}$$

Derivation

(b) Find the mean $\mu_X = E[X]$

Start with

$$E[X] = \int_0^\infty x\alpha e^{-\alpha x}\, dx.$$

Use integration by parts with

$$u = x, \quad dv = \alpha e^{-\alpha x} dx, \quad du = dx, \quad v = -e^{-\alpha x}.$$

Then

$$E[X] = \left[-xe^{-\alpha x}\right]_0^\infty + \int_0^\infty e^{-\alpha x}\, dx.$$

The boundary term is zero, and

$$\int_0^\infty e^{-\alpha x}\, dx = \left[-\frac{1}{\alpha} e^{-\alpha x}\right]_0^\infty = \frac{1}{\alpha}.$$

So

$$\mu_X = E[X] = \frac{1}{\alpha}.$$

Derivation

(c) Find the second moment $E[X^2]$

$$E[X^2] = \int_0^\infty x^2 \alpha e^{-\alpha x}\, dx.$$

Integrate by parts once:

$$u = x^2,\ dv = \alpha e^{-\alpha x} dx \Rightarrow du = 2x\, dx,\ v = -e^{-\alpha x}.$$

Hence

$$E[X^2] = \left[-x^2 e^{-\alpha x}\right]_0^\infty + 2\int_0^\infty x e^{-\alpha x}\, dx.$$

Define

$$I = \int_0^\infty x e^{-\alpha x}\, dx.$$

Integrate I by parts:

$$u = x,\ dv = e^{-\alpha x} dx \Rightarrow du = dx,\ v = -\frac{1}{\alpha} e^{-\alpha x}.$$

So

$$I = \left[-\frac{x}{\alpha} e^{-\alpha x}\right]_0^\infty + \frac{1}{\alpha}\int_0^\infty e^{-\alpha x} dx = 0 + \frac{1}{\alpha} \cdot \frac{1}{\alpha} = \frac{1}{\alpha^2}.$$

Therefore,

$$E[X^2] = 2I = \frac{2}{\alpha^2}.$$

Derivation

(d) Find the variance σ_X^2

Use

$$\sigma_X^2 = E[X^2] - (E[X])^2.$$

Substitute parts (b) and (c):

$$\sigma_X^2 = \frac{2}{\alpha^2} - \left(\frac{1}{\alpha}\right)^2 = \frac{1}{\alpha^2}.$$

Thus,

$$\sigma_X = \sqrt{\sigma_X^2} = \frac{1}{\alpha}.$$

Derivation

(e) Probability that X is within one standard deviation of the mean
From above,

$$\mu_X = \frac{1}{\alpha}, \qquad \sigma_X = \frac{1}{\alpha}.$$

So

$$\mu_X - \sigma_X = 0, \qquad \mu_X + \sigma_X = \frac{2}{\alpha}.$$

Hence

$$P(\mu_X - \sigma_X \le X \le \mu_X + \sigma_X) = P\left(0 \le X \le \frac{2}{\alpha}\right) = F_X\left(\frac{2}{\alpha}\right) - F_X(0).$$

Using the CDF,

$$F_X\left(\frac{2}{\alpha}\right) = 1 - e^{-2}, \qquad F_X(0) = 0.$$

Therefore

$$P\left(|X - \mu_X| \le \sigma_X\right) = 1 - e^{-2} \approx 0.8646647.$$

Final Result

$$F_X(x) = \begin{cases} 0, & x \le 0 \\ 1 - e^{-\alpha x}, & x > 0 \end{cases}, \quad E[X] = \frac{1}{\alpha}, \quad E[X^2] = \frac{2}{\alpha^2}, \quad \mathrm{Var}(X) = \frac{1}{\alpha^2},$$

$$P\left(|X - \mu_X| \le \sigma_X\right) = 1 - e^{-2} \approx 0.8647.$$

Exercise 2.8 (Page 76): Sum of Two Zero-Mean Uncorrelated RVs

Let W and V be zero-mean, uncorrelated RVs with standard deviations σ_W and σ_V. Define

$$X = W + V.$$

Derivation

Start from the definition of variance:

$$\mathrm{Var}(X) = E[(X - E[X])^2].$$

Because $E[W] = E[V] = 0$,

$$E[X] = E[W + V] = E[W] + E[V] = 0.$$

So

$$\mathrm{Var}(X) = E[(W + V)^2] = E[W^2 + 2WV + V^2] = E[W^2] + 2E[WV] + E[V^2].$$

For uncorrelated variables,

$$\mathrm{Cov}(W, V) = E[(W - E[W])(V - E[V])] = E[WV] = 0.$$

Hence

$$\mathrm{Var}(X) = E[W^2] + E[V^2] = \sigma_W^2 + \sigma_V^2.$$

Taking square root (standard deviation is nonnegative):

$$\sigma_X = \sqrt{\sigma_W^2 + \sigma_V^2}.$$

Final Result

$$\boxed{\sigma_X = \sqrt{\sigma_W^2 + \sigma_V^2}}$$

Exercise 2.12 (Page 76): $Y(t) = X\cos t$

Suppose X is a random variable and

$$Y(t) = X\cos t.$$

Derivation

(a) Expected value $E[Y(t)]$
For fixed t, $\cos t$ is deterministic, so it can be pulled out of expectation:

$$E[Y(t)] = E[X\cos t] = \cos t\, E[X].$$

If $\bar{x} := E[X]$, then

$$E[Y(t)] = \bar{x}\cos t.$$

Derivation

(b) Time average $A[Y(t)]$
For a particular realization x of X, the signal is $y(t) = x\cos t$. Its time average is

$$A[y(t)] = \lim_{T\to\infty} \frac{1}{T}\int_0^T x\cos t\, dt = x \lim_{T\to\infty} \frac{1}{T}\left[\sin t\right]_0^T = x \lim_{T\to\infty} \frac{\sin T}{T}.$$

Since $|\sin T| \leq 1$, we have $\sin T/T \to 0$, therefore

$$A[y(t)] = 0.$$

In shorthand notation this is often written as

$$A[Y(t)] = 0.$$

Derivation

(c) Condition under which $y(t) = A[Y(t)]$
Since $A[Y(t)] = 0$, the condition $y(t) = A[Y(t)]$ for all t becomes

$$x\cos t = 0 \quad \forall t.$$

This is true for all t only if

$$x = 0.$$

If the question is interpreted as equality between ensemble mean and time average, then

$$E[Y(t)] = A[Y(t)] \iff \bar{x}\cos t = 0\ \forall t \iff \bar{x} = 0.$$

Final Result

$$E[Y(t)] = \bar{x}\cos t, \qquad A[Y(t)] = 0.$$

For each realization, $y(t) = A[Y(t)]$ for all t only when $x = 0$. For mean/time-average equality for all t, the required condition is $E[X] = \bar{x} = 0$.